

Response to Referee R6

We thank the editor and referees for their continued engagement with our paper. Below we provide a brief summary of the main changes in this revision and then respond to each comment in turn.

- **Stock-return analysis.** [To be written — addresses R2’s main comment.]
- **Robustness with full set of fixed effects.** [To be written — addresses R6’s main comment on industry×year fixed effects.]
- **Limitations of the exposure measure.** [To be written — addresses R6’s comment on preferred shares.]
- **Pre-trends and event-time patterns.** [To be written — addresses R6’s comment on Figure 3.]
- **Liberty bond discussion.** We have revised the historical background section to clarify why Liberty Bond prices rose during the Supreme Court arguments in January 1935. Since Liberty Bonds were gold-denominated government obligations, a government loss would have obligated the Treasury to honor gold payments at the then-prevailing price of \$35 per ounce—69% above the pre-abrogation level—increasing the real value of the bonds to holders. We have also added the missing *New York Times* citation.

First, I appreciate the improvement in measuring d_{it} relative to all long-term obligations involving fixed contractual payments, but this only partly addresses my concerns. Preferred shares are not interest bearing, and thus may behave quite differently than debt during downturns or when firms are close to distress. I understand that there is nothing the author can do about this because bank debt is simply too small, but the paper should acknowledge this limitation more clearly and briefly discuss the potential confounders that cannot be addressed due to how firms structured their debt at that time.

The referee raises a valid point that we address more explicitly in the revised manuscript. Preferred shares are not interest-bearing: their dividends are discretionary and deferring them does not trigger default. This flexibility means that preferred shares behave quite differently from fixed-income obligations in distress situations. Because preferred shares appear in the denominator of $\tilde{d}$ but not the numerator, firms with a larger preferred share component have lower measured gold exposure. The concern is that high- $\tilde{d}$ firms—which rely more heavily on inflexible bond financing—may also be more generally vulnerable to economic downturns, independently of gold clause uncertainty. In that case, the investment differential we document in 1933–1934 could partly reflect a general debt-rigidity effect rather than the specific overhang from gold clause litigation.

We take several steps to address this concern, and we now discuss them explicitly in the manuscript. First, our preferred share placebo test (column 6 of Table 4) directly tests the alternative explanation: when we replace the numerator of $\tilde{d}$ with preferred shares (holding the denominator fixed), the 1933 and 1934 coefficients are positive and statistically insignificant. If the effect were driven by the general flexibility difference between preferred-heavy and bond-heavy firms, we would expect a negative effect here as well—firms with more preferred shares in the numerator would have lower measured exposure and thus higher investment. The absence of any effect is inconsistent with this story. Second, the temporal pattern of coefficients in Figure 6 is difficult to reconcile with a general debt-rigidity explanation: the effect of $\tilde{d}$ is concentrated precisely in 1933–1934 and reverses after the 1935 Supreme Court ruling, whereas a mechanism based on the rigidity of bond financing would predict persistent effects whenever economic conditions deteriorate (as in 1937–

1938, where we observe no differential effect). Third, the industry-year fixed effects in column 1 of Table 7 absorb time-varying shocks that may differentially affect firms by industry and liability composition, and the characteristic-interaction specifications in columns 2–10 allow leverage, size, and other observables that correlate with liability structure to follow their own time paths.

We nonetheless acknowledge that these tests do not fully eliminate the concern. An ideal research design would isolate variation in gold clause exposure while holding the debt-versus-preferred composition constant, but the institutional structure of the period—where virtually all corporate bonds had gold clauses and preferred shares had none—means this variation cannot be cleanly separated from liability composition. We have added a paragraph to the limitations discussion in Section 5 acknowledging this constraint explicitly.

I appreciate using 1936–37 as a placebo. While this helps address concerns about downturns driving the results, there are other important macro differences between both periods—such as the collapse of bond issuance, that limited rollover—that may affect exposed firms more in 1933–34 beyond the uncertainty of the abrogation of the gold clause. That is, the placebo helps but only for a subset of the concerns about confounders. To that end, the robustness regressions are more helpful. Though the findings are broadly reassuring, they do point to either the thinness of the data or a lack of robustness. The authors cannot control for industry $\times$ year and $X_{it} \times$ year at the same time, so the fixed effects vary across specifications in Table 6. The paper argues this is done “To assess robustness to firm-level controls separately from industry-level controls, this specification omits industry-year fixed effects.” This sounds like a copout. Given the concerns of New Deal policies as confounders, which may have affected firms differently given their characteristics, one may want to control for both. My sense is that they lack power, as even without the industry-year FEs the significance is lost for $1934 \times d$ across several specifications for Columns 2–10. I would like to see the results with industry-year FEs for these columns as well. If significance is lost across the board, then the paper should acknowledge it and make its best case to defend their preferred specifications due to limited power—but readers should know whether the effects are sensitive so they can pass their own judgement.

We thank the referee for these comments and address each in turn.

Placebo and the rollover channel. The referee correctly notes that the 1936–37 placebo addresses recession-driven confounders but not the collapse of bond issuance that may have disproportionately affected gold-exposed firms—which held more bonds—through a rollover channel. This concern is addressed directly by column 3 of Table 4, which excludes firms with bonds maturing at any point between 1931 and 1934. The 1933 and 1934 coefficients remain -0.058 and -0.037 , essentially unchanged from the baseline, indicating that the investment effect is not driven by firms with immediate refinancing needs.

Industry-year fixed effects in columns 2–10. We agree that readers should be able to assess the sensitivity of the results to this choice directly. We have re-run columns 2–10 of Table 7 (Table 6 in the previous version of the manuscript) replacing year fixed effects with industry-year fixed effects, so that all ten columns share the same fixed-effect structure as column 1. The results are reported in Internet Appendix Table IA.19, whose columns (1)–(9) correspond to columns 2–10 of Table 7.

The paper’s main result rests on the baseline specification of Table 4, which is already confirmed to be robust to industry-year fixed effects by column 1 of Table 7: the 1933 and 1934 coefficients are -0.059^{***} and -0.036^{**} , nearly identical to the baseline. The specifications in columns 2–10 of Table 7 are supplementary robustness checks for observable firm heterogeneity, not the preferred specification. Among those supplementary checks, adding industry-year fixed effects leaves $\tilde{d}$ negative and significant at the 5% level in all nine columns, and $1933 \times \tilde{d}$ significant in eight of nine.

The referee is correct that $1934 \times \tilde{d}$ loses significance in six of nine columns when industry-year fixed effects are added. We want to be clear, however, that this is a power problem rather than a robustness failure. The point estimates for $1934 \times \tilde{d}$ are negative and consistent in magnitude across all nine specifications, ranging from -0.011 to -0.036 , with no sign reversals or attenuation. It is the standard errors that widen, not the coefficients that shrink. This pattern is characteristic of reduced identifying variation, not of a coefficient that disappears under scrutiny.

The precision loss has a mechanical explanation. In the eight portfolio decile specifications, the

decile indicators are constructed from 1930 firm characteristics, and within a 2-digit industry firms tend to cluster in similar deciles; the decile-period dummies and industry-year cells therefore absorb overlapping variation in investment, widening standard errors. Because $1934 \times \tilde{d}$ has a smaller point estimate than $1933 \times \tilde{d}$, this widening is enough to push the 1934 interaction below conventional thresholds in six of these columns, while the 1933 interaction remains significant in all eight. The all-controls linear specification (column 2 of Table 7, shown as column (1) of Table IA.19) is the single column in which the 1933 interaction also loses significance, and it is the most demanding of the ten: it interacts all eight firm characteristics with the four-period structure simultaneously. In this specification the eight after-period interactions are collinear with the treatment timing and are dropped even under year fixed effects—this is a property of the baseline column 2 of Table 7, not something induced by the industry-year cells—so that layering industry-year fixed effects on top leaves little residual within-industry variation and both the 1933 and 1934 interactions become imprecise. In every column the point estimates retain their sign and magnitude; it is their standard errors that grow.

Column 1 of Table 7 remains our preferred specification for addressing the New Deal confounder concern. Industry-year fixed effects absorb any shock that affected an industry uniformly in a given year, which covers the canonical channels through which New Deal policies influenced investment. The characteristic controls in columns 2–10 address a distinct concern—that gold exposure may proxy for observable firm characteristics that followed differential investment trajectories—and they continue to provide reassuring evidence even when industry-year fixed effects are added, since $\tilde{d}$ remains significant throughout. We acknowledge that combining both robustness exercises simultaneously comes at a power cost, discuss this limitation explicitly in Section 5, and provide IA Table IA.19 so that readers can form their own judgment.

The authors made no comment about the fact that the 1933 and 1934 coefficients do not seem statistically different than the 1930 effects (Figure 3).

We acknowledge this observation, and we now address it in the revised manuscript. Two points bear on it. First, the identifying contrast in our design is not between the litigation years and

1930, but between each litigation year and the omitted base year of 1932—the year immediately preceding the abrogation. Relative to 1932, the 1933 and 1934 interaction coefficients are individually significant, whereas none of the pre-treatment coefficients, including 1930, is. The relevant difference-in-differences comparison therefore holds; requiring 1933 to be statistically distinguishable from a separately and noisily estimated 1930 coefficient is not the contrast that identifies the effect. Second, the confidence intervals around individual year estimates are wide given two-way clustered standard errors in a sample of this size, so pairwise comparisons between any two individual years will often fail to reach significance.

The correct test for parallel trends is not whether each pre-treatment year can be individually distinguished from 1933 or 1934, but whether the pre-treatment coefficients are jointly zero relative to the omitted category (1932) and whether they exhibit a systematic trend leading into the treatment period. The pre-treatment coefficients in column 2 of Table 4 are +0.026, +0.012, +0.022, −0.032, and −0.007 for 1927 through 1931, respectively, none of which is statistically significant. Crucially, these coefficients are jointly small and individually insignificant, and the two years immediately preceding the litigation—1931 (−0.007) and the omitted 1932 (0.000)—sit essentially at zero, so there is no trend running into the treatment window that could account for the sharp 1933 effect.

The 1930 dip (−0.032, insignificant) deserves a brief comment. In 1930 the Great Depression was already deepening—bank failures were accelerating and credit conditions were tightening sharply. Since gold-exposed firms were systematically more leveraged (Table 2), they may have been somewhat more sensitive to this general credit contraction than unexposed firms, independent of gold clause exposure. The fact that this coefficient is small and insignificant, and is followed by coefficients of −0.007 and 0.000 (omitted) in 1931–1932, is more consistent with occasional noise than with a persistent pre-trend.

That said, we acknowledge that limited statistical power prevents us from rejecting the null that the 1930 coefficient equals the 1933 or 1934 coefficient. We have added a sentence to Section 5 noting that the parallel trends argument rests on the full pre-treatment pattern rather than year-by-year pairwise tests, and that the absence of a visible trend in the point estimates provides the primary

support for this assumption.

One small comment—my understanding (though I could be wrong) is that repurchases in the open market were not allowed post 1929. I imagine the main reason driving the decline in shares was reorganizations. If that is the case, it is not obvious one would want to include this if the point of the analysis is to estimate whether firms were using payout policy to preemptively give the cash to shareholders. That said, the results in Table 4 show that the effects are driven by dividends, not repurchases, so this is mostly about guiding modern readers on what firm policies were in the 1930s.

The referee is essentially correct, and we are glad to clarify this for modern readers. Deliberate open-market repurchases were not a meaningful payout vehicle for industrial firms in this period. In practice, the changes in shares outstanding that our data capture over 1930–1934 primarily reflect mechanical share retirements arising from reorganizations, mandatory sinking fund payments, and call provisions—contractually required or involuntary reductions rather than discretionary cash distributions to shareholders.

We agree that modern readers would benefit from this context, and we have added a footnote to Section 5 clarifying the nature of “net repurchases” in the 1930s and noting that this component of total payout largely captures mechanical share retirements rather than deliberate equity distributions.

This does not affect any of our substantive conclusions. As the referee notes, Table 5 shows that the payout effect is driven entirely by cash dividends (column 2 is significant), while the net repurchase coefficient (column 3) is statistically insignificant in 1933 and 1934. The channel relevant to our debt overhang interpretation—shareholders extracting value through discretionary distributions in anticipation of losing their claim on the assets—operates through dividends, not through these involuntary share retirements.

Additional comment (cover letter)

A key issue is whether the authors can clearly disentangle the effects of uncertainty about the

abrogation of gold clauses from other confounders—other policies and firm characteristics, and whether it is about the gold clause vs interest-bearing leverage. There is not much they can do about the second given how firms structured their debt. I appreciate the refinement in the definition of the treatment effect, but the ideal case would be not to have preferred shares either. I am willing to look passed this, but the paper should be more upfront. The bigger issue is other confounders. They have tried to address this in the robustness section in Table 6, but as you can see they cannot control for a full set of fixed effects. What bothers me more is that they have not shown us the results with all effects, and instead state they do not include industry-year FEs in all columns to assess the robustness to industry and firm separately. This is a weak argument. If the issue is that they have too few firms to do this without the treatment effects being killed, they should say that. Readers may understand, but they need to know. And I would like to see how bad effects get—is it noisy or do the results become small and insignificant? I also find the placebo less helpful than the paper claims, but that’s a personal taste.

We are grateful for the referee’s overall assessment and for the clarity of this summary. We agree with the central message—that the paper should be upfront about what the data can and cannot establish—and we have revised the manuscript accordingly. We respond to the two main issues in turn, and indicate where each is now addressed in the paper.

Gold clause exposure versus interest-bearing leverage (preferred shares). The referee is correct that the ideal design would isolate gold-clause exposure while holding the debt-versus-preferred composition fixed, and that the institutional structure of the period does not permit this: virtually all corporate bonds of the era carried gold clauses, while preferred shares carried none. We no longer leave this implicit. Section 5.5 now contains an explicit limitations paragraph stating that $\tilde{d}$ is necessarily entangled with the composition of a firm’s fixed claims, that preferred-heavy firms may behave differently in distress for reasons unrelated to gold clauses, and that our tests mitigate but cannot fully eliminate this concern. We retain the preferred-share placebo (column 6 of Table 4), which finds no investment effect when the numerator of $\tilde{d}$ is replaced with preferred shares, as the most direct evidence that the effect is not driven by the general flexibility difference between

bond-heavy and preferred-heavy firms.

Other confounders and the full set of fixed effects. We agree that the previous draft’s justification for not always including industry-year fixed effects read as a rationalization, and we have removed that framing. We now do two things. First, we state the real reason plainly in Section 5.5: with a sample of this size, absorbing industry-year variation and characteristic-year variation simultaneously exhausts the identifying variation available to estimate the treatment effects. Second, and most importantly, we now show the results. Internet Appendix Table IA.19 re-estimates all of columns 2–10 of Table 7 with industry-year fixed effects, so the reader sees exactly “how bad effects get.” The answer is that they become noisier, not small: across all nine specifications the average exposure effect $\tilde{d}$ remains negative and significant at the 5% level, $1933 \times \tilde{d}$ remains significant in eight of nine, and the $1934 \times \tilde{d}$ point estimates remain negative and stable in magnitude (-0.011 to -0.036 , no sign reversals) even where they lose significance. This is the signature of reduced power rather than a disappearing effect, and we now say so directly rather than asking the reader to take it on faith.

On the placebo. We take the referee’s point that the 1936–37 placebo addresses only a subset of confounders—chiefly recession-driven ones—and not, for example, the collapse of bond issuance and the attendant rollover risk. We have tempered the paper’s language about the placebo accordingly and now lean more heavily on the direct robustness checks, including the exclusion of firms with bonds maturing during the litigation window (column 3 of Table 4), which speaks specifically to the rollover channel.

Taken together, these revisions are intended to make the paper’s limitations visible to readers while preserving what we believe is the core, robust finding: gold-exposed firms reduced investment specifically during the 1933–1934 litigation window and reversed afterward, a pattern that survives industry-year fixed effects in our preferred specification and that is not reproduced by the preferred-share or bank-debt placebos.

REFERENCES